

Exposé

Comparative Evaluation of the XGBoost Boosters *gbtree* and *gblinear*

In Terms of Predictive Accuracy and Efficiency for Binary User Classification

1 Topic and Relevance

Machine Learning plays an important role in digital user behavior analysis. In this context, XGBoost is a widely adopted algorithm due to its high accuracy and speed. There are two different booster types of which *gbtree* and *gblinear* are commonly used. However, these two boosters differ in their internal logic and computational behavior. This project aims to compare them using the same data and settings to evaluate which one performs better in a binary classification task. The comparison is relevant for Bigdata applications where performance and speed both matters.

2 Research Question and Hypothesis

Research Question:

Which booster type (*gbtree* or *gblinear*) performs better in terms of predictive accuracy and efficiency when used for binary classification?

Hypothesis:

The *gbtree* booster will achieve better results, while *gblinear* may require less computation time.

3 Practical Case

The analysis is based on structured tabular data representing customer profiles. The data contains typical customer-related features and a binary target variable indicating whether a customer is an app user or not. The models will be trained to predict this status, based on profile characteristics.

Beyond mere classification, the predicted probability (i.e., model confidence) is intended to serve as a propensity score in business context. This score reflects how likely a given instance belongs to the "user" class and can be used, for example, to prioritize marketing activities or personalize recommendations.

4 Methodology

- Dataset: Pre-processed CSV data with numerical, categorical and Boolean features
- Preprocessing: Encoding of categorical values, cleaning, normalization if needed
- Model Training:
 - Two models: XGBoost with *gbtree*, and XGBoost with *gblinear*
 - Standard hyperparameters, same evaluation setup
- Evaluation:
 - 5-fold cross-validation
 - ML Accuracy Metrics (Accuracy, F1-score, Precision, Recall, ROC-AUC)
 - Time: Training time and inference time
- Tools: Python, Jupyter Notebook, scikit-learn, xgboost

5 Required Theoretical Background and Literature

The project is based on supervised learning theory and boosting algorithms. Both booster types will be explained, along with the evaluation metrics.

Initial sources:

Chen, T. & Guestrin, C. (2016): *XGBoost: A Scalable Tree Boosting System*. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*, pp. 785–794. <https://doi.org/10.1145/2939672.2939785>

Osisanwo, F.Y. et al. (2017): *Supervised machine learning algorithms: classification and comparison*. In: *International Journal of Computer Trends and Technology (IJCTT)*, 48(3), pp. 128–138.

Vanhove, A.J. et al. (2025): *Classification Performance of Supervised Machine Learning to Predict Human Resource Management Outcomes: A Meta-Analysis Using Cross-Classified Multilevel Modeling*. In: *Human Resource Management*. <https://doi.org/10.1002/hrm.70012>

Kirasich, K., Smith, T. & Sadler, B. (2018): *Random forest vs logistic regression: binary classification for heterogeneous datasets*. In: *SMU Data Science Review*, 1(3), Article 9.

Natekin, A. & Knoll, A. (2013): *Gradient boosting machines, a tutorial*. In: *Frontiers in Neurorobotics*, 7, Article 21. <https://doi.org/10.3389/fnbot.2013.00021>

6 Draft Structure

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